

FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2022
FULL TEST – I
PAPER –2
TEST DATE: 23-12-2021

Time Allotted: 3 Hours

Maximum Marks: 180

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts**.
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections: Section-A, Section-B & Section-C**.
Section – A (01 –06, 20 – 25, 39 – 44): This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
Section – A (07 –10, 26 – 29, 45 – 48): This section contains **SIX (06) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
Section – B (11 – 13, 30 – 32, 49 – 51): This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.
Section – C (14 – 19, 33 – 38, 52 – 57): This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

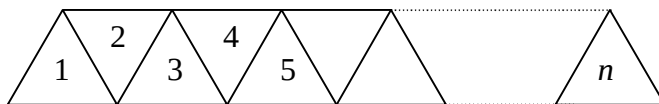
Physics

PART – I

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

1. n number of identical equilateral prisms are kept in contact as shown in figure. If deviation through a single prism is δ . Then (n, m are integers)

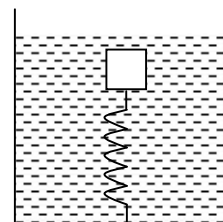


- (A) if $n = 2m$, deviation through n prism is zero.
- (B) if $n = 2m + 1$, deviation through system of n prisms is δ .
- (C) if $n = 2m$, deviation through system of n prism is δ
- (D) if $n = 2m + 1$ deviation through system of n prisms is zero

Ans. A, B

Sol. The consecutive prism produce equal and opposite deviation. Hence if number of prism is even deviation is zero and if odd, deviation is δ .

2. A block of mass ' m ' is attached by means of a spring to the bottom of a tank of water as shown in figure. At equilibrium, the spring is under compression. If the tank is now allowed to fall freely, then choose the correct alternative(s)



- (A) the spring comes to its relaxed position
- (B) the spring compresses more than its equilibrium compression
- (C) the buoyant force becomes zero
- (D) there will be some elongation in the spring

Ans. B, C

Sol. As the tank allowed to fall vertically, so buoyant force becomes zero, so the spring compresses more than its equilibrium compression.

3. A 10 m long horizontal stainless steel wire AB of mass 1 kg whose end A is fixed, is connected to a massless string BC passing over a smooth pulley. String BC is connected to a container of mass 2 kg at end C . Water (density = $1 \times 10^3 \text{ kg/m}^3$) is poured in the container at a constant rate of 2.25 litre/sec at $t = 0$. Also at $t = 0$, a pulse is generated at end A .
- (A) Time taken by the pulse to reach point B is 0.612 sec
 - (B) Time taken by the pulse to reach point B is 0.212 sec

(C) Tension in the string at this moment is 33.77 N

(D) Tension in the string at this moment is 24.77 N

Ans. A, C

Sol. Mass per unit length of the wire = 0.1 kg/m.
At any instant, tension in the string is $(2 + 2.25t)$ g N

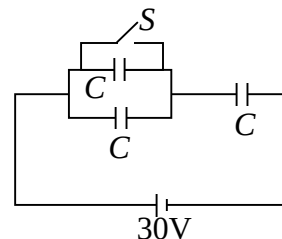
$$v = \sqrt{\frac{(2 + 2.25t)g}{0.1}}$$

$$\int_0^{10} dx = 10 \int_0^t \sqrt{2 + 2.25t} dt$$

$$t = 0.612 \text{ s}$$

$$T = (2 + 2.25 \times 0.612) \times 10 = 33.77 \text{ N}$$

4. Three capacitors each having capacitance $C = 2\mu\text{F}$ are connected with a battery of emf 30V as shown in the figure. When the switch S is closed.



(A) the amount of charge flown through the battery is $20 \mu\text{C}$

(B) the heat generated in the circuit is 0.6 mJ

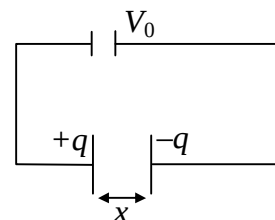
(C) the energy supplied by the battery is 0.6 mJ

(D) the amount of charge flown through the switch S is $60 \mu\text{C}$

Ans. A, C, D

Sol. Charge flow through battery = $20 \mu\text{C}$
 U = energy supplied by battery = 0.6 mJ
 U_i = energy store in capacitor's before closing = 0.6 mJ
 U_f = energy store in capacitor after closing = 0.9 mJ
 Heat generated = $U - (U_f - U_i) = 0.3 \text{ mJ}$, charge flow through switch is $60 \mu\text{C}$.

5. Two plates of a parallel plate capacitors carry charges q and $-q$ and are separated by a distance x from each other. The capacitor is connected to a constant voltage source V_0 . The distance between the plates is changed to $x + dx$. Then in steady state.



(A) change in electrostatic energy stored in the capacitor is $\frac{-Udx}{x}$, where U is initial potential energy.

- (B) change in electrostatic energy stored in the capacitor is $-\frac{Ux}{dx}$
- (C) attraction force between the plates is $\frac{1}{2}qE$
- (D) attraction force between the plates is qE . (where E is electric field between the plates)

Ans. A, C

Sol. Initial stored energy $U_i = \frac{1}{2} \frac{\epsilon_0 A V_0^2}{x}$

Final stored energy $U_f = \frac{1}{2} \frac{\epsilon_0 A V_0^2}{(x + dx)}$

$$\begin{aligned} \text{So, } \Delta U &= U_f - U_i = \frac{1}{2} \epsilon_0 A V_0^2 \left[\frac{1}{x + dx} - \frac{1}{x} \right] \\ &= \frac{1}{2} \epsilon_0 A V_0^2 \left[\frac{x - x - dx}{x(x + dx)} \right] = -\frac{1}{2} \frac{\epsilon_0 A V_0^2}{x^2} dx = -\frac{1}{2} \frac{\epsilon_0 A V_0^2}{x} \cdot \frac{dx}{x} = -\frac{U dx}{x} \end{aligned}$$

So, choice (a) is correct and choice (b) is wrong. $F = -\frac{dU}{dx} = -\frac{U}{x}$

$$F = -\frac{1}{2} \frac{\epsilon_0 A}{x} \cdot \frac{V_0^2}{x} = -\frac{1}{2} \left(\frac{\epsilon_0 A}{x} \cdot V_0 \right) \cdot \frac{V_0}{x} = -\frac{1}{2} qE$$

So magnitude of attractive force is $\frac{1}{2}qE$.

So, choice (c) is correct and choice (d) is wrong.

6. A charged particle of unit mass and unit charge moves with velocity of $\vec{v} = (8\hat{i} + 6\hat{j})\text{m/s}$ in a magnetic field of $\vec{B} = 2\hat{k}T$. Choose the correct alternative(s)

- (A) The path of the particle may be $x^2 + y^2 - 4x - 21 = 0$
- (B) The path of the particle may be $x^2 + y^2 = 25$
- (C) The path of the particle may be $y^2 + z^2 = 25$
- (D) The time period of the particle will be 3.14s

Ans. A, B, D

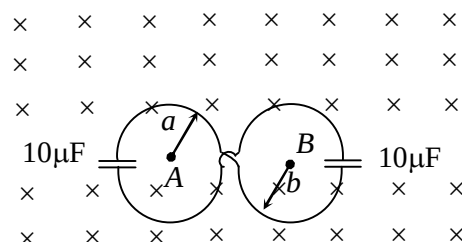
Sol. Motion is in x-y plane and $T = \frac{2\pi m}{qB} = \frac{2\pi(1)}{(1)(2)} = \pi\text{s}$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 07 and 08

A plane loop is shaped in the form, as shown in the figure with radii $a = 20$ cm and $b = 10$ cm and is placed in a uniform time varying magnetic field $B = (20 + 10t)$ T, where t is the time in second. Answer the following questions based on the above statement



7. The induced emf in the loop is

- (A) 0.942 V
- (B) 1.57 V
- (C) 0.157 V
- (D) 0.0942 V

Ans. A

Sol. ϕ = instantaneous flux through loop = $BA_1 + BA_2$

Sense of induced emf is opposite in each loop

$$\text{Hence, the net emf induced} = A_1 \frac{dB}{dt} - A_2 \frac{dB}{dt} = 10\pi(a^2 - b^2) = 0.942 \text{ V}$$

8. The maximum charge on each capacitor is

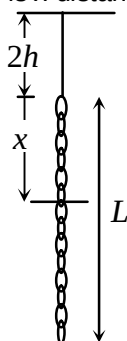
- (A) 4.71×10^{-6} C
- (B) 0.471×10^{-6} C
- (C) 52×10^{-6} C
- (D) 5.2×10^{-6} C

Ans. A

Sol. $Q = CV = 5 \times 10^{-6} \times 0.942 = 4.71 \times 10^{-6}$ C

Paragraph for Question Nos. 09 and 10

A uniform chain of length L is suspended in equilibrium by an elastic string of natural length h as shown. The chain is then suddenly cut at point A which is x distance below the upper end of chain.



9. If $x = L/3$, the velocity of suspended part of chain when it rises by $\frac{h}{2}$ from its equilibrium position is

(A) $\sqrt{\frac{gh}{3}}$

(B) $\sqrt{\frac{4}{3}gh}$

(C) $\sqrt{\frac{5}{4}gh}$

(D) $\sqrt{\frac{2}{3}gh}$

Ans. C

Sol. For equilibrium $kh = mg$
 $x = L/3$ suspended mass = $m/3$

$$\frac{1}{2} \frac{m}{3} v^2 + \frac{mg}{3} \frac{h}{2} + \frac{1}{2} k \left(\frac{h}{2} \right)^2 = \frac{1}{2} kh^2$$

$$v = \sqrt{\frac{5}{4}gh}$$

10. To allow the elastic string to become slack, the minimum value of x is

(A) $\frac{L}{2}$

(B) $\frac{L}{3}$

(C) $\frac{L}{4}$

(D) $\frac{2L}{3}$

Ans. A

 Sol. To just slack height raised = h

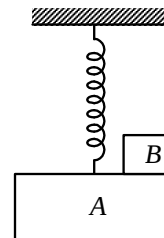
$$\frac{mx}{L}gh = \frac{1}{2}kh^2$$

$$x = \frac{L}{2}$$

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

11. A block A of mass 2 kg is hanging in a vertical plane with a spring of stiffness constant $k = 100 \text{ N/m}$. A block B of mass 1 kg is kept on block A and the system is in equilibrium. Suddenly block B is removed. The amplitude of resulting SHM of A is



Ans. 10

$$\text{Sol. Shift of equilibrium position} = \frac{3g}{k} - \frac{2g}{k} = \frac{g}{k} = \frac{10}{100} = 0.1 \text{ m} = 10 \text{ cm}$$

Hence amplitude = 10 cm

12. A plano-convex lens has a thickness of 4 cm. When placed on a horizontal table with curved surface in contact with it, the apparent depth of the bottom most point of the lens is found to be 3 cm. If the lens is inverted such that the plane face is in contact with the table, the apparent depth of the centre of plane face is found to be $25/8 \text{ cm}$. The focal length of the lens is

Ans. 75

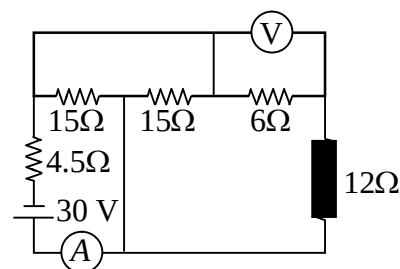
$$\text{Sol. } 3 = \frac{4}{\mu} \Rightarrow \mu = \frac{4}{3}$$

$$R = 25 \text{ cm}$$

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$f = 75 \text{ cm}$$

13. A galvanometer of coil resistance 1Ω is converted into voltmeter by using a resistance of 5Ω in series and same galvanometer is converted into ammeter by using a shunt of 1Ω . Now ammeter and voltmeter connected in circuit as shown, find the reading of voltmeter.



Ans. 3

Sol. $R_{\text{Voltmeter}} = 6\Omega$, $R_{\text{ammeter}} = 0.5\Omega$

$$R_{\text{eq}} = 10\Omega$$

$$I = \frac{30}{10} = 3A$$

$$\text{Reading of voltmeter} = 1 \times 3 = 3 \text{ volt.}$$

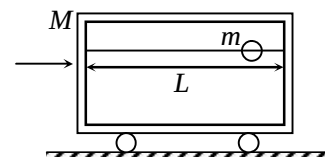
Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 14 and 15

Question Stem

A horizontal frictionless string is threaded through a bead of mass m . The string is kept in tension between two vertical opposite sides of a cart of mass M (see figure). Length of thread is L and radius of bead is r ($r \ll L$). Initially, the bead is at right edge of the cart. At $t = 0$, the cart is imparted a velocity v_0 . All collisions are elastic. [$v_0 = 10\text{m/s}$, $M = 2\text{kg}$, $m = 1\text{kg}$, $L = 1\text{m}$]



14. The velocity of C.O.M of the cart and the bead after 3 successive collisions is

Ans. 6.67

Sol.
$$V_{\text{CM}} = \frac{Mv_0 + m(0)}{M + m}$$

15. Distance traveled by cart before second collision is

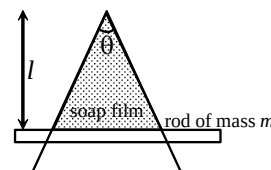
Ans. 1.33

Sol.
$$\begin{aligned} \text{Distance} &= L + \left(\frac{M - m}{m + M} \right) v_0 \frac{L}{v_0} \\ &= \left(\frac{2M}{M + m} \right) L \end{aligned}$$

Question Stem for Question Nos. 16 and 17

Question Stem

A wire is bent at an angle θ . A rod of mass m can slide along the bend wire without friction as shown in figure. If a soap film is maintained in the frame and frame is kept in a vertical position, the rod is in equilibrium. [$\theta = 90^\circ$, $m = 100\text{gm}$, $l = 1\text{m}$]



16. Surface tension of the soap solution is [in S.I system]

Ans. 0.25

Sol. $2T \left(2l \tan \frac{\theta}{2} \right) = mg, T = \frac{mg}{4l \tan \frac{\theta}{2}}$

17. If rod is displaced slightly in vertical direction, find the time period of small oscillation [take $g^{1/2} = \pi$]

Ans. 2.00

Sol. If x be the displacement in vertical direction of the rod from equilibrium position

$$F_{\text{res}} = -2x \tan \frac{\theta}{2} T \times 2, \quad a = \frac{-4T \tan \frac{\theta}{2}}{m} x,$$

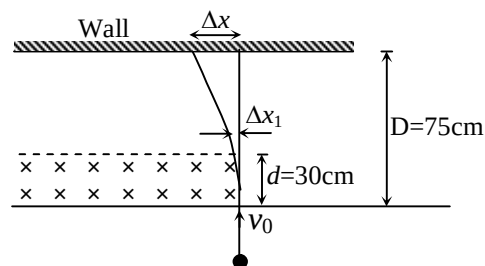
$$T = 2\pi \sqrt{\frac{m}{4T \tan \frac{\theta}{2}}} = 2\pi \sqrt{\frac{m}{4 \frac{mg}{4l}}} = 2\pi \sqrt{\frac{l}{g}}$$

Question Stem for Question Nos. 18 and 19

Question Stem

The magnetic force on a moving charge is given by $\vec{F} = q\vec{v} \times \vec{B}$. Further when a charge particle moves in uniform magnetic field such that its velocity is perpendicular to the magnetic field then its path is circular.

A particle with charge $2.4\mu\text{C}$ and mass $3.2 \times 10^{-12} \text{ kg}$ is initially traveling in the $+y$ direction with a speed $v_0 = 1.35 \times 10^5 \text{ m/s}$. It then enters a region containing a

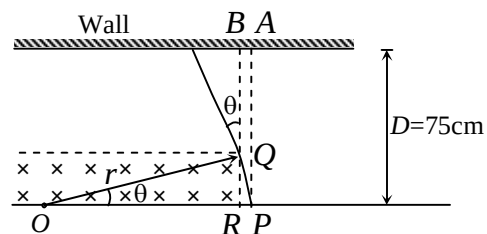


uniform magnetic field that is directed into, and perpendicular to, the page. The magnitude of field is 0.360 T . The region extends a distance of 30 cm along the initial direction of travel; 75 cm from the point of entry in to the magnetic field region there is a wall. The length of the field free region is thus 45 cm . When the charged particle enters the magnetic field, it follows a curved path whose radius of curvature is r . It then leaves the magnetic field after time t_1 , having been deflected a distance Δx_1 . The particle then travels in the field-free region and strikes the wall after undergoing a total deflection Δx .

18. The radius r of the curved part of the path is(in cm)

Ans. 50.00

Sol.
$$R = \frac{mv}{qB} = \frac{3.2 \times 10^{-12} \times 1.35 \times 10^5}{2.4 \times 10^{-6} \times 0.36} = 0.5 \text{ m}$$



19. The total horizontal deflection is(in cm)

Ans. 44.00

Sol.
$$\Delta x = AB + BC = \Delta x_1 + BQ \tan \theta = \Delta x_1 + (D - d) \tan \theta = 0.1 + (0.75 - 0.30) \frac{3}{4} = 0.44 \text{ m}$$

Chemistry

PART – II

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

20. The time required for an electron to make one complete revolution around nucleus of H-atom in a higher orbit n_2 is 8 times to that of a lower orbit n_1 . Therefore n_1 and n_2 are

- (A) 1 and 2
- (B) 2 and 3
- (C) 2 and 4
- (D) 3 and 6

Ans. A, C, D

Sol. Time period $\propto n^3$

$$\frac{T_1}{T_2} = \left(\frac{n_1}{n_2}\right)^3$$

$$T_2 = 8 T_1 \Rightarrow n_2 = 2n_1$$

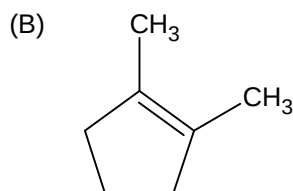
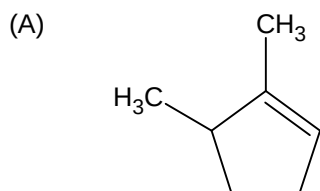
21. Which of the following compounds acts both an oxidising as well as a reducing agent?

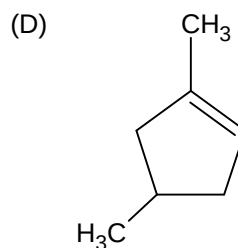
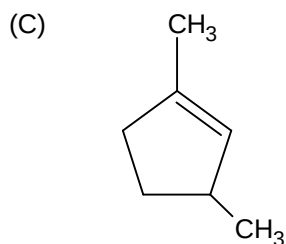
- (A) HNO_2
- (B) H_2O_2
- (C) H_2S
- (D) SO_2

Ans. A, B, D

Sol. In H_2S oxidation number of sulphur is minimum-2, it can act as only reducing agent.

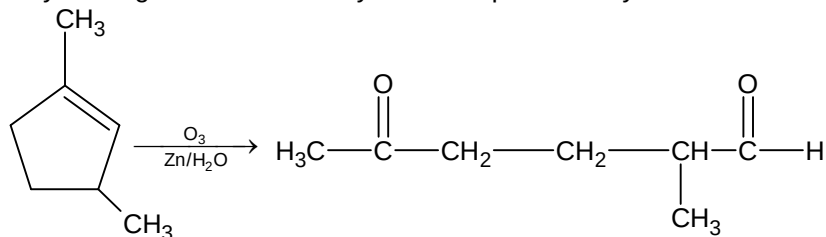
22. Which compounds would not give 5-keto-2-methylhexanal upon ozonolysis?



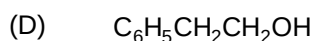
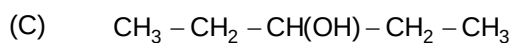
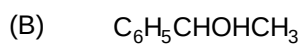
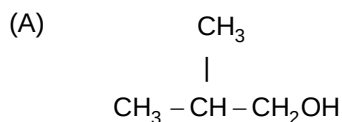


Ans. A, B, D

Sol. Only C will give 5-keto-2-methylhexanal upon ozonolysis.



23. Among the follow compounds which do not give positive iodoform test



Ans. A, C, D

Sol. Alcohols lacking βCH_3 group do not give iodoform test.

24. Metals which form protective layer of oxide when react with dil. HNO_3



Ans. A, B

Sol. Al, Cr form protective layer of oxide and renders passive.

- Ans. A, C, D

$\text{Ni}^{2+} \Rightarrow [\text{Ar}]$  

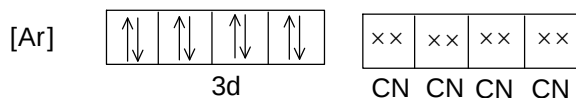
⇒ [Ar]

↑↓	↑↓	↑↓	↑↓
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xx	xx	xx	xx
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3d CN CN CN CN

$$\text{Ni} \Rightarrow [\text{Ar}]3d^8 4s^2$$



$$\text{EAN} = [28 - 0 + 2 \times 4] = 36$$

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

A charged particle like electron produces induced magnetic field when it is in motion. Therefore the revolving electrons of an atom can create magnetic field. The paramagnetic substances are attracted towards an external magnetic field because they possess certain magnetic moment. The magnetic moment of an atom depends on the number of unpaired electrons. It is determined by using the formula

$$\mu = \sqrt{n(n+2)} \text{B.M}$$

26. The magnetic moment of an oxide of vanadium is 1.732 B.M. What is the molecular formula of the oxide?

- (A) VO
(B) VO₂
(C) V₂O
(D) V₂O₃

Ans. B

Sol. $\sqrt{n(n+2)} = 1.732 = \sqrt{3}$

or, $n(n+2) = 3$

or, $n = 1$

The electronic configuration of vanadium is:

$V = [Ar]_{18} 3d^3 4s^2$

In V^{4+} state, vanadium will contain one unpaired electron.

∴ The molecular formula of its oxide is V₂O₄ or VO₂

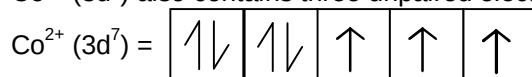
27. The magnetic moment of which of the following ions is same as that of nitrogen atom?

- (A) Co²⁺
(B) Fe³⁺
(C) Ni²⁺
(D) Zn²⁺

Ans. A

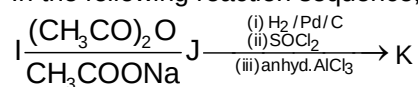
Sol. Nitrogen contains three unpaired electrons.

Co²⁺ (3d⁷) also contains three unpaired electrons



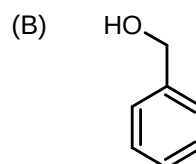
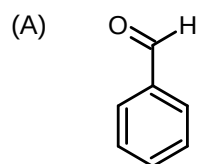
Paragraph for Question Nos. 28 and 29

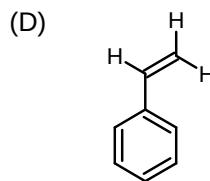
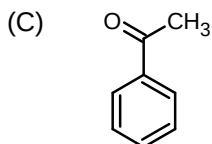
In the following reaction sequence, the compound J is an intermediate



J(C₉H₈O₂) gives effervescence on treatment with NaHCO₃ and a positive Baeyer's test

28. The compound I is

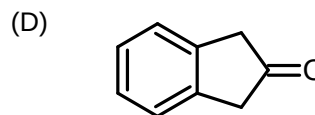
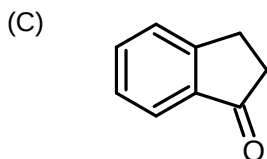
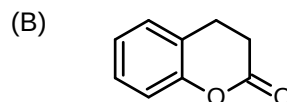
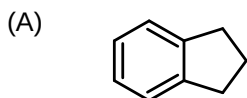




Ans. A

Sol. Perkin reaction

29. The compound K is



Ans. C

Sol. FC acylation

Section – B (Maximum Marks: 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.

30. The trade name of one stereoisomer of compound B is Aspartame. How many total stereoisomers of Compound B exist?

Ans. 4

Sol. There are two chiral centre so total stereoisomers will be 4.

31. 1.3245 g of a monobasic acid when dissolved in 100 g of water lowers the freezing point by 0.2046°C . 0.2 g of the same acid when dissolved and titrated required $15.1 \text{ ml of } \frac{N}{10}$ alkali. (Assuming molarity = molality), the pH of acid solution is: (K_f of water $1.86 \text{ K kg mol}^{-1}$)

Ans. 2

Sol. Meq. of alkali = Meq. of acid

$$\Rightarrow 15.1 \times \frac{1}{10} = \frac{0.2}{M} \times 1000$$

On, solving $M = 132.45 \text{ g mol}^{-1}$

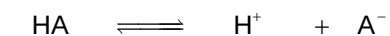
$$\text{Now, molality of acid solution (m)} = \frac{1.3245 \times 1000}{132.45 \times 100} = 0.1$$

$$\therefore \Delta T_f = i \times K_f \times m$$

$$0.2048 = i \times 1.86 \times 0.1$$

$$I = 1.1 = 1 + \alpha$$

$$\therefore \alpha = 0.1$$



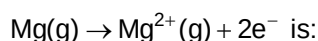
$$\text{Now, } C \quad \quad \quad 0 \quad \quad \quad 0$$

$$C - C\alpha \quad \quad \quad C - C\alpha \quad \quad \quad C\alpha$$

$$[\text{H}^+] = C\alpha = 0.1 \times 0.1 = 10^{-2}$$

$$\therefore \text{pH} = 2$$

32. I.E.₁ and I.E.₂ of Mg are 178 and 348 kcal mol⁻¹. The enthalpy required for the reaction in kcal/mole



Ans. 526

Sol. I.E. = I.E.₁ + I.E.₂

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 33 and 34

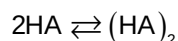
Question Stem

On addition of 0.25 mole of a non volatile solute HA (mono basic acid) in 5 moles of benzene solute gets partially dimerised and the pressure of solution measured is 581.11 mm Hg.

On further titration of this solution 200 ml of 1.25 M NaOH required to reach end point.

$$(\rho_{\text{NaOH}} = 1.21 \text{ gm/ml})$$

$$\left\{ \begin{array}{l} P_{\text{Benzene}}^{\circ} = 600 \text{ mm Hg} \\ P_{\text{H}_2\text{O}}^{\circ} = 23.76 \text{ mm Hg} \end{array} \right\}$$



33. Degree of association of solute (α) in solution is?

Ans. 0.70

34. How many mL of 1.25M NaOH will be required for complete neutralization of the acid?

Ans. 200.00

Sol. (Q.33 – 34)

$$P_s = P^{\circ} \cdot X_{\text{solvent}}$$

$$\frac{P^{\circ} - P_s}{P_s} = \frac{n_{\text{solute}}}{N_{\text{solvent}}}$$

$$\frac{600 - 581.11}{581.11} = \frac{n}{5}, n = 0.1625$$

$\alpha = 0.7$

Question Stem for Question Nos. 37 and 38

Question Stem

Internal energy is a state function quantity. It is a measurement of total energy of system. For ideal gas potential energy is zero therefore internal energy is function of temperature only. For ideal gas molecules kinetic energy is divided in three part?

(i) Translational (ii) Rotational (iii) Vibrational

vibrational energy become effective only at very high temperature

Take $R = 2 \text{ cal/mol/K}$

37. For diatomic ideal gas molecules C_p (molar heat capacity) value (in cal) is equal to

Ans. 7.00

38. C_p value (in cal) for water molecules in vapour state is equal to (assuming ideal condition)

Ans. 8.00

Sol. (Q.37-38)

$$C_v = (\text{degrees of freedom}) \times \frac{R}{2}$$

Mathematics

PART – III

Section – A (Maximum Marks: 24)

This section contains **SIX (06)** question. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).

39. The number of solutions in $[0, \pi]$ of the equation

$$\sin^4 x + \cos^6 x = a \text{ is}$$

- (A) none if $a < \frac{1}{4}$
- (B) three if $a = 1$
- (C) two if $a = \frac{1}{4}$
- (D) three if $a = \frac{5}{4}$

Ans. A, B, C

Sol. $a = 1 - \frac{3}{8}(1 - \cos 4x) \Rightarrow \cos 4x = \frac{8a - 5}{3}$

For real solution

$$-1 \leq \frac{8a - 5}{3} \leq 1$$

$$-3 = 8a - 5 \leq 3$$

$$2 \leq 8a \leq 8 \Rightarrow \frac{1}{4} \leq a \leq 1$$

40. Which of the following statement(s) is/are correct?

- (A) The number of quadratic equations having real roots which remain unchanged even after squaring their roots is 3.
- (B) The number of solutions of the equation $\tan 2\theta + \tan 3\theta = 0$, in the interval $[0, \pi]$ is equal to 6.
- (C) For $x_1, x_2, x_3 > 0$ the minimum value of $\frac{2x_1}{x_2} + \frac{128x_3^2}{x_2^2} + \frac{x_2^3}{4x_1x_3^2}$ equals 24.
- (D) The locus of the mid points of chords of the circle $x^2 + y^2 - 2x - 6y - 1 = 0$, which are passing through origin is $x^2 + y^2 - x - 3y = 0$.

Ans. A, B, D

Sol. (A) We have $\alpha + \beta = \alpha^2 + \beta^2$ and $\alpha\beta = \alpha^2\beta^2$

$\therefore$ Number of quadratic equations with real roots are 3.

$$(B) \tan 3\theta = \tan(\pi - 2\theta) \Rightarrow 3\theta = n\pi + (\pi - 2\theta), n \in \mathbb{Z}$$

$$\therefore \theta = (n+1)\frac{\pi}{5}, n \in \mathbb{I},$$

$$\text{Putting } n = -1, 0, 1, 2, 3, 4$$

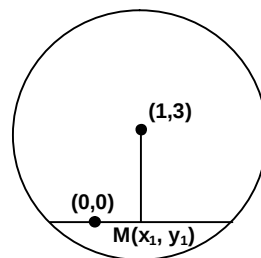
$$\text{We get } \theta = 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \frac{5\pi}{5}$$

$$(C) AM \geq GM \Rightarrow \text{minimum value} = 4$$

$$(D) \text{ Let } M(x_1, y_1) \text{ be the mid-point of chord}$$

$$\text{Then } m_1 m_2 = -1 \Rightarrow \left(\frac{y_1 - 3}{x_1 - 1} \right) \left(\frac{y_1}{x_1} \right) = -1$$

$$\text{Locus of } M(x_1, y_1) \text{ is } x^2 + y^2 - x - 3y = 0$$



41. If $4 \sin \alpha \cos \beta + 2 \cos \beta - \sqrt{3} = 2\sqrt{3} \sin \alpha$, where $\alpha, \beta \in [0, 2\pi]$ and $L = \alpha - \beta$ then

(A) maximum value of L is $\frac{11\pi}{6}$

(B) minimum value of L is $-\frac{5\pi}{6}$

(C) maximum value of L is $\frac{5\pi}{3}$

(D) minimum value of L is $-\frac{11\pi}{6}$

Ans. A, D

Sol. $(2 \cos \beta - \sqrt{3})(2 \sin \alpha + 1) = 0$

$$\therefore \cos \beta = \frac{\sqrt{3}}{2} \text{ or } \sin \alpha = -\frac{1}{2}$$

$$\beta = \frac{\pi}{6}, \frac{11\pi}{6} \text{ or } \alpha = \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$\therefore L = \alpha - \beta = \begin{cases} \text{maximum where } \alpha = \frac{11\pi}{6} \text{ and } \beta = 0 \\ \text{minimum when } \alpha = 0, \beta = \frac{11\pi}{6} \end{cases}$$

$$L_{\max} = 11\frac{\pi}{6}; L_{\min} = -\frac{11\pi}{6}$$

42. A differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the functional equation

$$f(x).f(y) + f(x+y) = e^x f(y) + e^y f(x) + xy \quad \forall x, y \in \mathbb{R}. \text{ If } f'(0) = 0 \text{ and } f(0) = 0, \text{ then which of the following statements is/are correct ?}$$

(A) $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \frac{1}{2}$

$$(B) \quad \int_x^{x^2} (f'(t) - f(t)) dt > 0 \quad \forall |x| > 1$$

$$(C) \quad F(x_2) > F(x_1) \quad \forall x_2 > x_1, \text{ where } F(x) = f'(x) - f(x)$$

$$(D) \quad \text{There exists at least two horizontal tangents to the curve } y = f(x) \text{ in } (-1, 1).$$

Ans. A, B, C

$$\text{Sol. } f(x).f(y) + f(x+y) = e^x f(y) + e^y f(x) + xy.$$

$$\therefore f(x).f'(y) + f'(x+y) = e^x f'(y) + e^y f'(x) + x$$

Put $y = 0$

$$f(x).f'(0) + f'(x) = e^x f'(0) + f(x) + x$$

$$\Rightarrow f'(x) - f(x) = x \Rightarrow f(x) = e^x - x - 1$$

$$\therefore (A) \quad \lim_{x \rightarrow 0} \frac{f(x)}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2} = \frac{1}{2}.$$

$$(B) \quad \int_x^{x^2} t dt = \left(\frac{t^2}{2} \right)_x^{x^2} = \frac{x^4 - x^2}{2} > 0 \quad \forall |x| > 1$$

$$(C) \quad F(x) = f'(x) - f(x) = (e^x - 1) - (e^x - x - 1) = x \text{ which is increasing function so}$$

$$F(x_2) > F(x_1) \text{ if } x_2 > x_1$$

$$(D) \quad \text{Now } f'(x) = e^x - 1 \text{ and } f''(x) = e^x > 0 \quad \forall x \in \mathbb{R}$$

$$\therefore f'(x) \text{ is increasing function on } \mathbb{R}$$

$$\therefore f'(-1) = -ve \text{ and } f'(1) = +ve$$

$$\therefore f'(c) = 0 \text{ has exactly one root in } (-1, 1).$$

i.e., one horizontal tangent. In $(-1, 1)$

43. Let $f(x)$ be the remainder when $p(x) = x^{73} - 2x^6 + 3x^3$ is divided by $x^4(x^2 - 1)$. Identify which of the following statement is(are) correct?

$$(A) \quad \text{Range of } f(f(x)) \text{ is } \mathbb{R}$$

$$(B) \quad \text{Range of } f(f(x)) \text{ is } [0, \infty)$$

$$(C) \quad f(x) \text{ has exactly 3 points of inflection.}$$

$$(D) \quad f(x) \text{ has exactly one point of inflection.}$$

Ans. A, D

$$\text{Sol. } P(x) = x^{73} - 2x^6 + 3x^3 = Q(x).x^4(x^2 - 1) + (ax^2 + bx + c)x^3$$

After cancelling x^3

$$\text{Put } x = 0 \Rightarrow c = 3.$$

$$x = 1 \Rightarrow a + b + c = 2$$

$$x = -1 \Rightarrow a - b + c = 6$$

$$\therefore a = 1, b = -2$$

$$\therefore f(x) = x^5 - 2x^4 + 3x^3$$

Degree of $f(f(x))$ is 25 so range of $f(f(x))$ is $\mathbb{R}$

$$f''(x) = 20x^3 - 24x^2 + 18x$$

Only one real root

44. Let $S_1 : x^2 + y^2 - 10x + 16 = 0$ and S_2 be image of S_1 w.r.t line $y = x$ and S_3 be image of S_2 w.r.t $y = 0$. If S' be the circle which cuts all three circles orthogonally, then

- (A) $S' : x^2 + y^2 = 16$
 (B) $S_2 : x^2 + y^2 - 10y + 16 = 0$
 (C) $S_3 : x^2 + y^2 + 10y + 16 = 0$
 (D) radius of director circle of S' is $8\sqrt{2}$

Ans. A, B, C

Sol. $\therefore$ Centre of $S_1 = (5, 0)$ and radius $r_1 = 3$
 $\therefore$ Centre of $S_2 = (0, 5)$ and radius $r_2 = 3$ and centre of $S_3 = (0, -5)$ and radius $r_3 = 3$.
 $\therefore$ Radical centre of S_1, S_2, S_3 will be $(0, 0)$
 Length of tangent from $(0, 0)$ upon S_1 or S_2 or $S_3 = 4$
 $\therefore S' : x^2 + y^2 = 16$

Section – A (Maximum Marks: 12)

This section contains **TWO (02) paragraphs**. Based on each paragraph, there are **TWO (02) questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.

Paragraph for Question Nos. 45 and 46

Let Π_1 denotes the equation of the plane to which the vector $(1, 1, 0)$ is normal and which contains the line L whose equation is $\vec{r} = \hat{i} + \hat{j} + \hat{k} + \lambda(\hat{i} - \hat{j} - \hat{k})$. Π_2 denotes the equation of the plane containing the line L and a point with position vector $(0, 1, 0)$.

45. Vector of magnitude $\sqrt{6}$ along the line of intersection of planes Π_1 and Π_2 perpendicular to normal vector of plane Π_1 , is
- (A) $\pm\sqrt{2}(-1, 1, 1)$
 (B) $\pm\sqrt{2}(1, -1, 1)$
 (C) $\pm\sqrt{2}(1, 1, -1)$
 (D) $\pm\sqrt{2}(1, 1, 1)$

Ans. A

46. The acute angle between Π_1 and Π_2 is

- (A) $\tan^{-1}(1) - \tan^{-1}(\sqrt{2} - 1)$
 (B) $\tan^{-1}(1) - \tan^{-1}(2 - \sqrt{3})$
 (C) $\tan^{-1}(1) - \tan^{-1}(\sqrt{2} + 1)$
 (D) $\tan^{-1}(1) - \tan^{-1}(2 + \sqrt{3})$

Ans. B

Sol. (Q.45 & Q46)

$$\vec{r} \cdot \vec{n}_1 = \vec{a} \cdot \vec{n}$$

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (\hat{i} + \hat{j}) = 2$$

$$\Rightarrow x + y = 2$$

.... (Plane Π_1)

Now, $\vec{AB} = \hat{i} + \hat{k}$; A(1,1,1) and B(0, 1, 0)

$$\text{Now, } \vec{AB} \times \vec{V} = \vec{n}_2, \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 1 \\ 1 & -1 & -1 \end{vmatrix}$$

$$\vec{n}_2 = \hat{i}(0+1) - \hat{j}(-1-1) + \hat{k}(-1)$$

$$\Rightarrow \vec{n}_2 = \hat{i} + 2\hat{j} - \hat{k}$$

Hence, equation of plane

$$\Pi_2 \text{ is } (\vec{r} - \hat{j}) \cdot (\hat{i} + 2\hat{j} - \hat{k}) = 0$$

$$\text{Or } x + 2y - z = 2$$

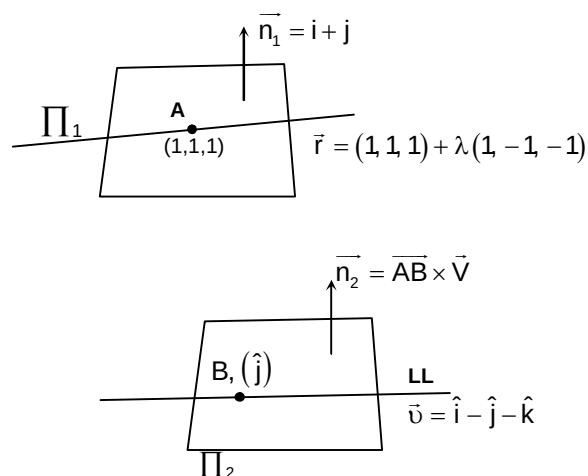
(i) Vector along the line of intersection of planes

Π_1 and Π_2 is $\vec{n}_1 \times \vec{n}_2$ which is also perpendicular to the normal vector of plane Π_1 .

$$\therefore \vec{n}_1 \times \vec{n}_2 = -\hat{i} + \hat{j} + \hat{k}, \text{ so required vector} = \pm \sqrt{6} \left(\frac{-\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}} \right)$$

(ii) If θ is the acute angle between Π_1 and Π_2 then

$$\cos \theta = \frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} = \frac{(\hat{i} + \hat{j}) \cdot (\hat{i} + 2\hat{j} - \hat{k})}{\sqrt{2} \sqrt{6}} = \frac{(1+2)}{2\sqrt{3}} = \frac{\sqrt{3}}{2} \Rightarrow \theta = \frac{\pi}{6}$$



Paragraph for Question Nos. 47 and 48

Let A be a square matrix of order $n \times n$. A constant λ is said to be characteristic root of A if there exists a $n \times 1$ matrix X such that $AX = \lambda X$

47. If 0 is a characteristic root of A, then

- (A) A is non-singular
 (B) A is singular

(C) $A = O$

(D) $A = I_n$

Ans. B

Sol. If $\lambda = 0$, we get $|A| = 0$
 $\Rightarrow A$ is singular.

48. If λ is a characteristic root of A and $n \in \mathbb{N}$, then λ^n is a characteristic root of

(A) A^n

(B) A^{n-1}

(C) A^{-n}

(D) $A - A^n + A^{-n}$

Ans. A

Sol. We have, $A^2X = A(AX) = A(\lambda X)$
 $= \lambda(AX) = \lambda^2X$
 $A^3X = A(A^2X) = A(\lambda^2X)$
 $= \lambda^2(AX) = \lambda^2(\lambda X)$
 $= \lambda^3X$
 Continuing in this way, we obtain
 $A^nX = \lambda^nX \forall n \in \mathbb{N}$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**.*

49. If the plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(\alpha)$ with x-axis then the value of 7α is

Ans. 2

Sol. $\theta = \sin^{-1}(\alpha) \Rightarrow \sin \theta = \alpha$
 $\therefore \cos\left(\frac{\pi}{2} - \theta\right) = \frac{1.2 - 3.0 + 6.0}{\sqrt{4 + 9 + 36} \cdot \sqrt{1}}$
 $\sin \theta = \frac{2}{7} \Rightarrow \alpha = \frac{2}{7}$
 $\therefore 7\alpha = 2$

50. If $T_n = \frac{(2n+1)(2n^2+2n-1)}{(n-1)^2 n^2 (n+1)^2 (n+2)^2}$, $n \in \mathbb{N}, n \geq 2$. If $K = \lim_{n \rightarrow \infty} \sum_{r=2}^n T_r$ then $27K$ is equal to

Ans. 3

$$\begin{aligned}
 \text{Sol. } T_r &= \frac{(2r+1)(2r^2+2r-1)}{(r^2-1)^2(r^2+2r)^2} = \frac{(2r+1)(2r^2+2r-1)}{(r^2-1)^2((r+1)^2-1)^2} \\
 \therefore T_r &= \left[\frac{1}{(r^2-1)^2} - \frac{1}{((r+1)^2-1)^2} \right] \\
 \therefore K &= \lim_{n \rightarrow \infty} \sum \left(\left(\frac{1}{(2^2-1)^2} - \frac{1}{(3^2-1)^2} \right) + \left(\frac{1}{(3^2-1)^2} - \frac{1}{(4^2-1)^2} \right) + \dots + \left(\frac{1}{(n^2-1)^2} - \frac{1}{((n+1)^2-1)^2} \right) \right) \\
 \therefore k &= \frac{1}{9}
 \end{aligned}$$

51. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = x^3 - 3x^2 - 9x + 27$. If $x \in (a, b) \cup (c, d)$ then $f^3(x) - 3f^2(x) - 9f(x) + 27 < f(x^3 - 4x^2 - 3x + 19)$. Then $(b-a) + (c+d)$ is

Ans. 10

$$\begin{aligned}
 \text{Sol. } f'(x) &= 3(x^2 - 2x - 3) = 3(x+1)(x-3) \\
 \text{Case-I } f'(x) &> 0 \text{ i.e. } f(x) \text{ is increasing the } x > 3 \text{ or } x < -1 \\
 f(f(x)) &< f(x^3 - 4x^2 - 3x + 19) \Rightarrow f(x) < x^3 - 4x^2 - 3x + 19 \\
 \Rightarrow x^2 - 6x + 8 &< 0 \Rightarrow x \in (2, 4) \\
 \therefore x &\in (3, 4) \\
 \text{Case-II } f'(x) &< 0 \text{ i.e. } f(x) \text{ is decreasing then } -1 < x < 3 \\
 f(x) &> x^3 - 4x^2 - 3x + 19 \\
 \Rightarrow x^2 - 6x + 8 &> 0 \Rightarrow (x-2)(x-4) > 0 \Rightarrow x < 2 \text{ or } x > 4 \\
 \therefore x &\in (-1, 2) \\
 \therefore x &\in (-1, 2) \cup (3, 4) \\
 \therefore b - a + c + d &= 2 + 1 + 3 + 4 = 10
 \end{aligned}$$

Section – C (Maximum Marks: 12)

This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE**. If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

Question Stem for Question Nos. 52 and 53

Question Stem

Let f be a real valued function from $\mathbb{N}$ to $\mathbb{N}$ satisfying the relation $f(m+n) = f(m) + f(n)$ for all $m, n \in \mathbb{N}$.

52. If range of f contains all the even numbers, then the possible value of $f(1)$ are A & B then $\frac{3A+2B}{5}$ is equal to

Ans. 1.40

$$\begin{aligned}
 \text{Sol. } f(m+n) &= f(m) + f(n) \\
 m=1, n=N-1
 \end{aligned}$$

$$f(N) = f(1) + f(N-1)$$

$$f(N-1) = f(1) + f(N-2)$$

:

$$f(N) = Nf(1).$$

If $f(1) = 1 \Rightarrow f(N) = N$ which contains even numbers.

If $f(1) = 2 \Rightarrow f(N) = 2$ which contains even numbers.

53. If domain of f is first '3m' natural numbers and if the number of elements common in domain and range is 'm', then the value of $\frac{25f(1)}{4}$ is

Ans. 18.75

Sol. $f(n) = nf(1)$

Number of elements common in domain and range is m

$$n(R \cap D) = \frac{n(D)}{3}$$

Question Stem for Question Nos. 54 and 55

Question Stem

Let a sequence $T_1, T_2, T_3, \dots, T_n, \dots$ And $S_n = T_1 + T_2 + \dots + T_n$, $S'_n = S_1 + S_2 + S_3 + \dots + S_n$, $S''_n = S'_1 + S'_2 + S'_3 + \dots + S'_n$, $S''_n = n^4$

54. $\frac{T_3 - T_2}{4}$ is equal to

Ans. 5.75

Sol. $S' = 1, 15, 65, 175$

$S \equiv 1, 14, 50, 110$

$T \equiv 1, 13, 36, 60$

$T_n = 24n - 36$ for $n \geq 3$

55. T_n is given by $An + B$ then $\frac{(3A - B)}{5}$ is equal to

Ans. 21.60

Sol. $S' = 1, 15, 65, 175$

$S \equiv 1, 14, 50, 110$

$T \equiv 1, 13, 36, 60$

$T_n = 24n - 36$ for $n \geq 3$

Question Stem for Question Nos. 56 and 57

Question Stem

For a recursive relation in 'n' we may generate nth term free from any other term i.e. purely in 'n'. Based on this answer the following questions

56. If $(a_{n+1})^2 a_n + 2(a_{n+1}) - a_n = 0$ and $a_0 = 1$, then $\lim_{n \rightarrow \infty} 2^{n+4} (a_n)$ is

Ans. 12.56

Sol. Given equation converts to $a_n = \frac{2a_{n+1}}{1 - (a_{n+1})^2}$

$$\Rightarrow a_n = \tan\left(\frac{\pi}{2^{n+2}}\right)$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n 2^n = \frac{\pi}{4}$$

57. If $a_n = 2a_{n+1}^2 - 1$ and $a_0 = 0$, then $\lim_{n \rightarrow \infty} e^{a_n} \times 2$ (use $e = 2.72$)

Ans. 5.44

Sol. Given equation converts to $a_{n+1} = \sqrt{\frac{1+a_n}{2}}$

$$\Rightarrow a_n = \cos\left(\frac{\pi}{2^{n+1}}\right)$$

$$\text{Hence, } \lim_{n \rightarrow \infty} e^{\cos\left(\frac{\pi}{2^{n+1}}\right)} = e$$